

algorithms derived by means of a symmetric Trotter factorization of the Liouville propagator:

$$\begin{aligned} \exp[\tau L] &= \prod_{\alpha=1}^{n_s-1} \exp\left[\frac{\tau}{2}L_\alpha\right] \exp \exp[\tau L_{n_x}] \\ &\times \prod_{\beta=1}^{n_s-1} \exp\left[\frac{\tau}{2}L_{n_s-\beta}\right] \end{aligned} \tag{13}$$

are not only time-reversible but also measure-preserving.